MACHINE LEARNING WASTEWATER COST PREDICTION

Team Member Names:

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Sponsor Organization:

Brown & Caldwell and University of Louisville

Sponsor Names:

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BRIEF OVERVIEW OF PROJECT

The Machine Learning-Based Wastewater Cost Prediction System for Residential and Commercial Usage in Texas is a web-based application designed to analyze historical wastewater cost data from 2017-2025 to predict future wastewater fees for residential and commercial users across cities in Texas for the next 15 years. This information is important to the residents and the cities in regards to budgeting, infrastructure and rate planning. Our web based application will include filters, and prediction charts.

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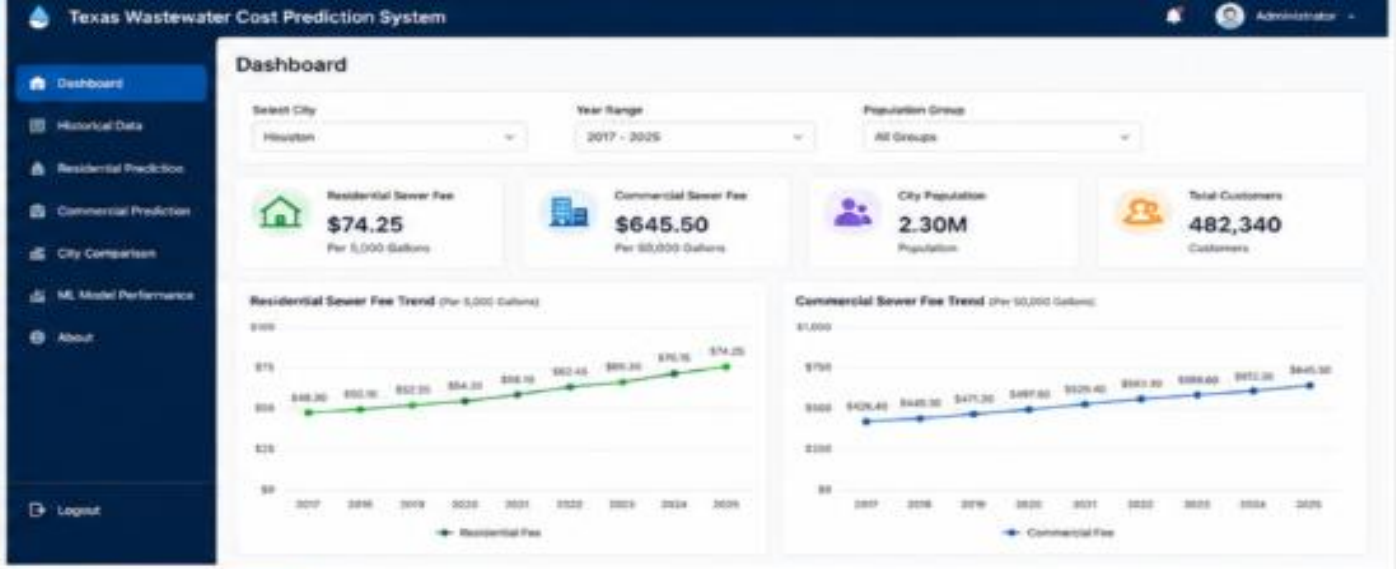
SYSTEM REQUIREMENTS

What does the system do?

1. **User Authentication:** Secure login/logout using username and password validation.
2. **Historical Data:** Store and retrieve wastewater data from 2017 to 2025 using a SQLite database.
3. **City/Year Filtering:** Search cities and filter wastewater data by year using frontend search and filtering functions.
4. **Trend Analysis**: Visualize residential and commercial wastewater cost trends using interactive charts.
5. **15-Year Prediction:** Predict future residential and commercial wastewater costs using trained machine learning models.
6. **ML Model Evaluation:** Compare Linear Regression, Random Forest, and XGBoost models using prediction accuracy metrics to select the best-performing model.
7. **Interactive Dashboard**: Display historical data, predictions, charts, and city comparisons through a React-based dashboard.

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TEXAS
WASTEWATER COST
PREDICTION SYSTEM
Machine Learning Based Analytics
Email / Username
Password
LOGIN
Wastewater Cost Analysis | 2017 – 2025


Texas Wastewater Cost Prediction System
Dashboard
Historical Data
Residential Prediction
Commercial Prediction
City Comparison
ML Model Performance
About
Logout
Dashboard
Select City
Houston
Year Range
2017 - 2025
Population Group
All Groups
Residential Sewer Fee
$74.25
Per 5,000 Gallons
Commercial Sewer Fee
$645.50
Per 50,000 Gallons
City Population
2.30M
Population
Total Customers
482,340
Customers
Residential Sewer Fee Trend (Per 5,000 Gallons)
Commercial Sewer Fee Trend (Per 50,000 Gallons)


Historical Data
Historical Wastewater Cost Data
Download CSV


Residential Sewer Fee Prediction (Per 5,000 Gallons)
Select City
Houston
Predict For
Next 15 Years (2026 - 2040)
Generate Prediction
Prediction Summary
$74.25
$94.80
$117.60
$145.30
95.67%


Commercial Sewer Fee Prediction (Per 50,000 Gallons)
Select City
Houston
Predict For
Next 15 Years (2026 - 2040)
Generate Prediction
Prediction Summary
$645.50
$812.40
$1,000.70
$1,248.30
93.44%


City Comparison
Select Metric
Residential Sewer Fee (Per 5,000 Gal)
Compare Cities
Houston
McKinney
Mesquite
Pasadena
McAllen
Compare
Comparison Insights
Lowest Cost (2025)
McAllen ($38.45)
Highest Cost (2025)
Houston ($74.25)
Fastest Growing Cost
McKinney (2.83%/yr)
Average Texas Fee (2025)
$58.71

INITIAL SYSTEMS SPECIFICATION

Functional requirements:

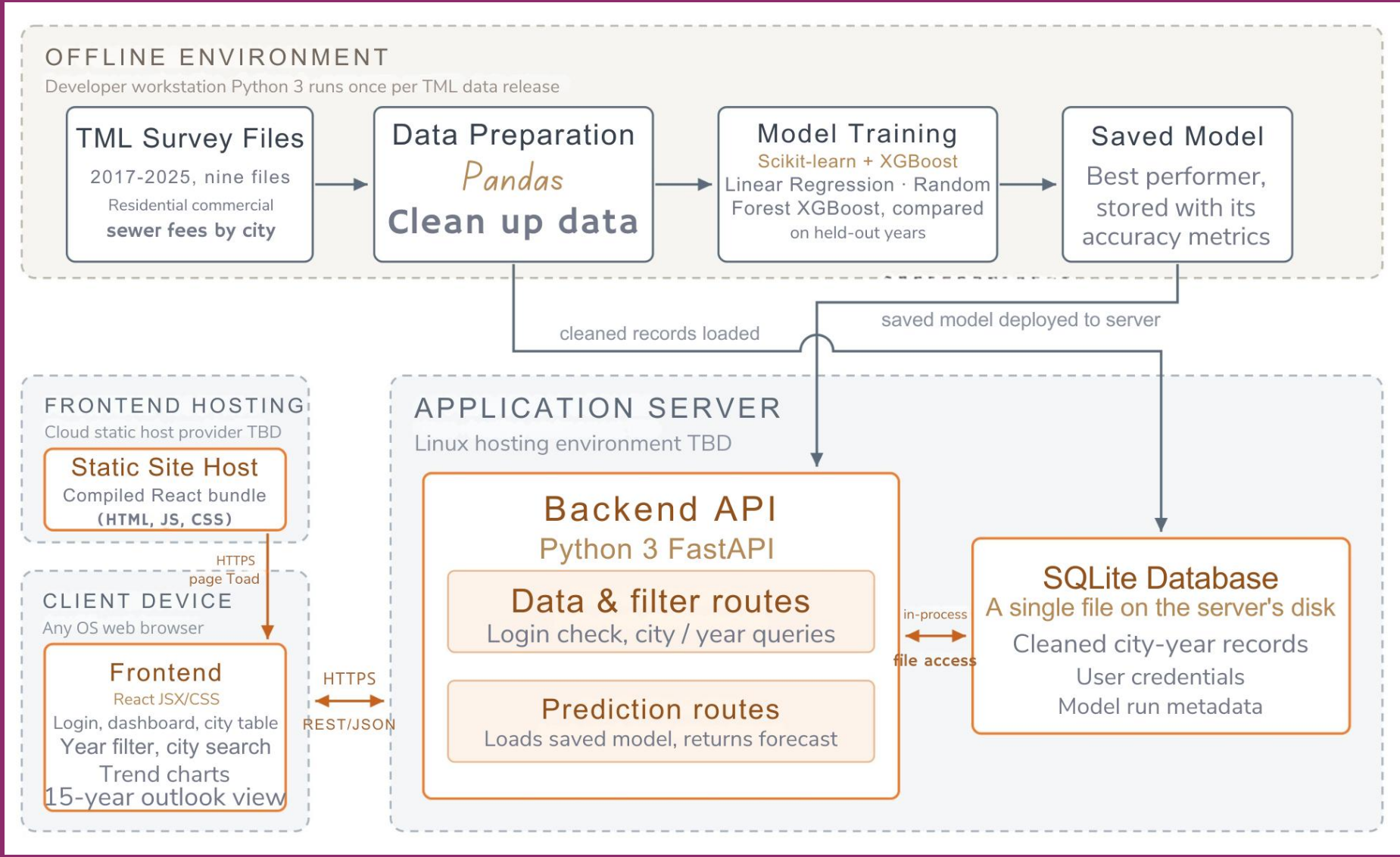
1.) User login: Allow the user to login securely in order to access the dashboard.
2.) Filter by city: Users will be able to search for the Texas city they want to access the wastewater data on.
3.) Historical data: The system will display the wastewater cost data from 2017-2025.
4.) Cost predictions: The system will generate future costs for wastewater in the residential and commercial settings for the next 15 years.

Non-Functional requirements:

1.) Usability: User friendly website.
2.) Scalability: The system should be able to handle the addition of more Texas cities, data or cost information.

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PROPOSED ARCHITECTURE



- Four run-time environments, offline through application server
- Model trained offline, deployed as a file
- Browser to backend: HTTPS / REST
- Backend to SQLite: in-process file access, same machine
- Deployed via GitHub Actions; hosts TBD

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IMPLEMENTATION MILESTONES

Data Preparation (Sept. 6-12)

- Import 2017-2025 wastewater data
- Clean data
- Organize residential & commercial records

Database Setup (Sept. 13-17)

- Create SQLite database
- Load cleaned city/year records
- Store user credentials & model metadata

Machine Learning Development (Sept. 18-27)

- Train Linear Regression
- Train Random Forest
- Train XGBoost
- Compare accuracy & select best model

Backend & Login (Sept. 28 - Oct. 6)

- Build Python/FastAPI backend
- Implement secure login/logout
- Create data and filtering API routes
- Connect backend to SQLite

Frontend & Dashboard (Sept. 18-27)

- Build React interface
- Create login & dashboard pages
- Add city search and year filters
- Display historical trend charts

Prediction Integration (Sept. 28 - Oct. 6)

- Deploy saved ML model
- Create prediction API route
- Generate 15-year forecasts
- Display residential/commercial predictions

System Testing (Oct. 7-14)

- Test frontend to backend communication
- Test database queries and filtering
- Test login & predictions
- Fix integration/usability issues

Deployment & Final Validation (Oct. 15-22)

- Deploy frontend & backend
- Configure database/server environments
- Verify all system requirements
- Prepare final demonstration

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TEAM ASSIGNMENTS & RESPONSIBILITIES

Quyen: Machine Learning

- Prepare data and train using Linear Regression, Random Forest, and XGBoost models
- Compare Model accuracy and select the best-performing model
- Generate the 15-year wastewater cost predictions

Emily: UI/UX

- Build the React-based user interface
- Create the login screen and dashboard layout
- Display historical data, trends, and prediction charts

Hannah: Backend & Login

- Develop the Python/FastAPI backend
- Implement secure login/logout and credential validation
- Create REST API endpoints for data and predictions

Adriana: Database & Integration

- Set up and maintain the SQLite database
- Develop city/year filtering queries
- Connect the frontend, backend, and database

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THANK YOU!

Questions?

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